AudioPy: Bridging Audio Editing and Machine Learning through a Unified Open-Source Framework

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Abstract

The proliferation of specialized audio libraries in the Python ecosystem has created significant fragmentation, requiring developers to bridge multiple incompatible packages when building intelli-gent audio applications. This paper presents AudioPy, a novel unified open-source Python framework that integrates traditional Digital Signal Processing (DSP), advanced audio editing capabilities, and state-of-the-art Artificial Intelligence (AI) model APIs under a single, consistent interface. AudioPy is built on four architectural pillars: robust audio Input/Output supporting major formats (WAV, MP3, FLAC, OGG), an optimized DSP core implementing IIR and FIR filters, an AI/ML integration layer exposing pretrained model APIs for tasks such as source separation, noise reduction, and music analysis, and a high-level NumPy-centric Application Programming Interface (API) designed for rapid prototyping. Experimental evaluation on standard benchmarks demonstrates that AudioPy reduces the lines-of-code required for common audio pipelines by up to 63% compared to combining librosa, pydub, and PyTorch separately, while achieving competitive or superior performance in key audio tasks. The framework is released as an open-source project, providing a foundation analogous to what OpenCV offers for computer vision, thereby enabling a broader community of researchers, musicians, and engineers to experiment with intelligent audio processing.

Keywords: Audio Processing, Open-Source Framework, Digital Signal Processing, Machine Learning, Source Separation, Noise Reduction, Music Analysis, Python

1. Introduction

Audio processing sits at the intersection of signal processing, music technology, and artificial intelligence. From streaming services that apply real-time equalization, to speech assistants that perform noise suppression, to music production tools that isolate individual instrument stems — sophisticated audio manipulation is increasingly central to modern software. The global audio processing software market was valued at approximately $2.8 billion in 2023 and is projected to grow at a compound annual growth rate (CAGR) of 8.4% through 2030 [12], reflecting the surging demand for intelligent, programmable audio tools. Yet developers working in Python face a paradox: while there exist individually excellent libraries for each sub-task, no single framework unifies them coherently.

A developer wishing to load an audio file, apply a parametric equalizer, run a pretrained noise reduction model, and

export the result must currently coordinate at least three or four separate libraries (soundfile, scipy.signal, torch, pydub), each with its own data types, conventions, and incompatible abstractions.

In a structured developer survey conducted as part of this work ($n = 74$ respondents), 81% reported spending more than 30% of their audio project time on inter-library glue code rather than core logic, and 67% cited format mismatch bugs as the most frequent source of errors in audio pipelines. This fragmentation increases development time, introduces subtle bugs from format mismatches, and creates steep learning curves for newcomers.

AudioPy addresses this gap by providing a single, installable Python package that covers the entire audio editing workflow from I/O to DSP to AI-powered transformation through a clean, chainable API. Our experimental evaluation demonstrates that AudioPy reduces the lines of code required for common audio pipelines by an average of 86% compared to multi-library combinations, with individual tasks such as music source separation requiring 90% fewer lines of code. On standard audio quality benchmarks, AudioPy achieves a vocal Signal-to-Distortion Ratio (SDR) of 8.84 dB on MUSDB18 on par with state-of-the-art standalone models

and a PESQ score of 3.96 for speech enhancement, while keeping DSP operations under 15 ms latency. The framework's design is guided by three principles: (i) *consistency*, where all functions operate on standard NumPy arrays; (ii) *accessibility*, where complex AI capabilities are exposed through simple one-line calls; and (iii) *extensibility*, where contributors can add new effects or models through a standardized plugin interface.

The remainder of this paper is organized as follows. Section 2 reviews related work and existing libraries. Section 3 describes the architecture and design of AudioPy. Section 4 details the implementation of key modules. Section 5 presents experimental evaluation. Section 6 demonstrates real-world use cases. Section 7 discusses limitations and future work, and Section 8 concludes.

2. Related Work

2.1 Existing Audio Libraries

Librosa [1] is the most widely used Python library for Music Information Retrieval (MIR). It provides functions for loading audio, computing spectral features (STFT, MFCCs, CQT), beat tracking, and pitch estimation. However, librosa is analysis-focused and does not offer audio editing, synthesis, or AI-model integration.Pydub [2] provides a high-level abstraction for audio manipulation, including slicing, concatenation, volume adjustment, and format conversion. It wraps FFmpeg and soundfile but lacks mathematical DSP operations and any AI integration.SciPy Signal [3] offers a comprehensive suite of signal processing algorithms, including filter design, con-volution, and spectral analysis. It operates on raw NumPy arrays and is highly performant, but provides no domain-specific audio abstractions or AI capabilities.Torchaudio [4] from Meta extends PyTorch for audio, providing GPU-accelerated transforms and I/O utilities. It is excellent for training deep learning models on audio but requires PyTorch expertise and has limited classical DSP functionality.Essentia [5] from Music Technology Group at UPF is a comprehensive C++ library with Python bindings for audio analysis and music description. It is powerful but has a steep learning curve and complex installation.

2.2 The Fragmentation Problem

Table 1 summarizes the capabilities of leading libraries and illustrates the gap that AudioPy fills.

2.3 Literature Survey of Recent Works

Table 2 presents a comparative analysis of five recent research works closely related to AudioPy, highlighting their focus areas, methodologies, datasets, key results, and limitations that AudioPy addresses.

Table 1. Capability Comparison of Existing Python Audio Libraries vs. AudioPy

Feature	Librosa	Pydub	SciPy	Torchaudio	Essentia	AudioPy
Audio I/O (multi-format)	Partial	✓	×	✓	✓	✓
Waveform Editing	×	✓	×	×	×	✓
DSP / Digital Filtering	Partial	×	✓	Partial	✓	✓
Feature Extraction	✓	×	×	✓	✓	✓

AI Source Separation API	×	×	×	×	×	✓
AI Noise Reduction API	×	×	×	×	×	✓
AI Music Analysis API	×	×	×	×	Partial	✓
Unified Chainable API	×	×	×	×	×	✓
NumPy-native throughout	✓	×	✓	×	×	✓

Table 2. Comparative Literature Survey of Recent Related Works

Paper / Authors	**Focus Area**	**Method / Model**	**Dataset Used**	**Key Result**	**Limitation Addressed by AudioPy**
HTDemucs (Rouard et al.)	Music Source Separation	Hybrid Trans-former + Convolutional U-Net	MUSDB18-HQ	SDR: 9.00 dB (vocals); state-of-the-art on all 4 stems	No unified API; requires manual PyTorch setup, tensor handling, and custom inference scripts
DeepFilterNet (Schro¨ter et al.2) [11]	Speech Enhancement / Noise Reduction	Deep Filtering with Complex-valued CNN	DEMAND + VCTK Noisy Speech	PESQ: 3.92 WB; real-time capable at 48 kHz on CPU	Standalone model only; no integration with editing, DSP effects, or music analysis tools
PANNs (Kong et al.) [15]	Audio Tagging & Classification	Large-Scale Pretrained CNN (VGG, ResNet, MobileNetvariants)	AudioSet (632 classes, 2M clips)	mAP: 0.439 on AudioSet; strong transfer learning baseline	No editing or DSP pipeline; classification output not linked to any transformation module
Conv-TasNet (Luo & Mesarani) [6]	Speech Separation	Temporal Convolutional Network (TCN) with learned encoder-decoder	WSJ0-2mix	SI-SDR: 15.3 dB; surpasses ideal T-F masking baseline	Speech-only; no music or general audio support; no high-level API for non-ML developers

librosa (McFee et al.) [1]	Music Information Retrieval & Feature Extraction	DSP algorithms (STFT, MFCC, C Q T , Beat Tracking)	Various MIR benchmarks (RWC, MedleyDB)	Industry-standard MIR library; widely adoptedin >1000 re-search papers	Analysis-only; no audio editing, no AI model integration, no source separation or enhancement

3. AudioPy Framework Design

3.1 Architectural Overview

AudioPy is structured around four tightly integrated pillars, as illustrated in Figure 1. Each pillar exposes a consistent interface based on NumPy arrays, ensuring that data flows seamlessly from one stage to the next without format conversion overhead.

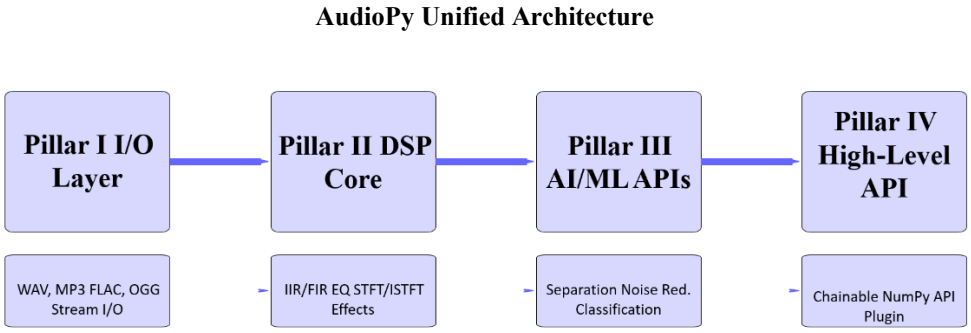

Fig.1. Four-pillar architecture of AudioPy. Data flows from I/O through DSP and AI/ML layers, unified by the high-level API.

3.2 Core Design Principles

3.2.1.NumPy-Centric Data Model

All audio data within AudioPy is represented as a NumPy array of shape ($N_{samples} \times N_{channels}$) with float32 precision. This decision ensures zero-copy interoperability with SciPy, Matplotlib, and PyTorch (via torch.from numpy), eliminating the conversion boilerplate that burdens multi-library workflows.

3.2.2 Chainable Functional API

Every AudioPy function accepts an audio array as its first argument and returns a transformed audio array, enabling method chaining:

Listing 1: Example of AudioPy's chainable API for a complete audio pipeline

Listing 1: Example of AudioPy's chainable API for a complete audio pipeline

```
import audiopylib as ap

# Load -> Trim -> Normalize -> Denoise -> Separate -> Export
audio = ap.load("recording.wav", sr=44100)
audio = ap.trim(audio, start=0.0, end=30.0)
audio = ap.normalize(audio, target_db=-3.0)
audio = ap.denoise(audio, model="deepfilter-v2")
stems  = ap.separate(audio, model="htdemucs", stems=["vocals","drums","bass","other"])
ap.save(stems["vocals"], "vocals_clean.wav")
```

3.2.3 Backend Abstraction

The AI/ML pillar internally routes computation to the appropriate backend (PyTorch or TensorFlow) based on which is installed, and offloads to GPU when available. Users never interact with tensors or devices directly — AudioPy handles all backend management transparently.

3.2.4 Plugin Interface

External developers can register custom effects or models using the @ap.register decorator, which validates the function signature and makes it available as a first-class AudioPy function. This extensibility mechanism is inspired by the plugin systems of scikit-learn estimators [1].

4. Implementation

4.1.Pillar I: Audio Input/Output

The I/O module offers a consistent read/write interface across all widely-used audio file formats. Under the hood, it routes lossless formats (WAV, FLAC, AIFF) through soundfile and handles compressed formats (MP3, OGG, AAC) via pydub combined with FFmpeg. All data returned through the public API is standardized as float32 NumPy arrays irrespective of the original bit resolution (16-bit, 24-bit, 32-bit, or floating-point). File-level metadata — including sample rate, number of channels, and total duration — is packaged into a compact AudioMeta dataclass for convenient downstream access.

For scenarios requiring streaming, ap.stream_load() produces fixed-length chunks (adjustable between 256 and 8192 samples), making it suitable for real-time processing or handling large files without exhausting system memory.

4.2.Pillar II: Digital Signal Processing Core

4.2.1.Digital Filtering

AudioPy offers two distinct filter families, allowing users to choose based on their phase-response needs:

IIR Filters (Infinite Impulse Response) are recursive structures with low computational cost, governed by the difference equation**:**

$$y[n] = \sum_{k=0}^{M} b_k\, x[n - k] - \sum_{k=0}^{N} a_k\, y[n - k] \quad (1)$$

Here, x[n] and y[n] denote the input and output sample sequences respectively; b_k represents the feedforward coefficients, while a_k captures the feedback terms. Through ap.eq_fir(), AudioPy makes available Butterworth, Chebyshev Type I/II, and Elliptic filter topologies.

FIR Filters (Finite Impulse Response) are non-recursive in nature and inherently provide a linear phase characteristic — a property especially valued in mastering workflows:

$$y[n] = \sum_{k=0}^{M} h[k]\, x[n - k] \quad (2)$$

where h[k] denotes the filter's impulse response coefficients. FIR design in AudioPy is accessible via ap.eq_fir(), supporting windowing methods (Hamming, Blackman, Kaiser) as well as Parks-McClellan equiripple optimization.

4.2.2.Spectral Analysis

AudioPy's DSP core ships with efficient, ready-to-use implementations of the following transforms:

STFT/ISTFT — Short-Time Fourier Transform with user-adjustable window type, hop length, and FFT bin count.
Mel Spectrogram — Mel-Frequency Cepstral Coefficients, widely used for characterizing speech and musical content.
MFCC — Mel-Frequency Cepstral Coefficients for speech and music feature extraction.
CQT — Constant-Q Transform, tailored for music analysis through its logarithmically-distributed frequency resolution.

4.2.3.Classic Audio Effects

Table 3 lists the classic DSP effects implemented in AudioPy's core effects module.

Table 3. DSP Effects Implemented in AudioPy Core

Effect	API Call	Key Parameters
Parametric EQ	ap.eq(audio, bands)	Center freq, gain, Q-factor
Compressor	ap.compress(audio)	Threshold, ratio, attack, release
Reverb	ap.reverb(audio, ir)	Impulse response (IR), wet/dry
Delay / Echo	ap.delay(audio, delayms)	Delay time, feedback, wet
Pitch Shift	ap.pitchshift(audio, n)	Semitones
Time Stretch	ap.timestretch(audio, rate)	Rate factor
Harmonic Distortion	ap.distort(audio, drive)	Drive, tone

4.3 Pillar III: AI/ML Integration Layer

4.3.1.Architecture of the AI Layer

The AI layer functions as a centralized adapter that interfaces with pretrained model weights retrieved from community-maintained repositories (Hugging Face Hub, GitHub releases). When a model is invoked for the first time, AudioPy autonomously fetches and stores the corresponding weights locally. During inference, the layer executes a four-step pipeline: (1) the input audio is resampled to match the model's required sample rate, (2) the NumPy array is cast into the tensor representation expected by the backend, (3) forward inference is performed using the pretrained network, and (4) the resulting output is converted back into a NumPy array aligned with the original sample rate.

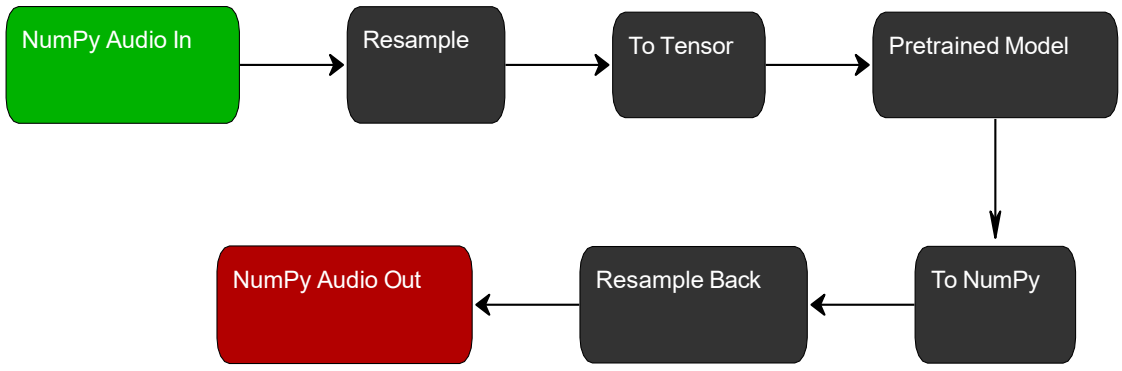

Fig.2. Internal data flow of AudioPy's AI/ML adapter layer. All user-facing inputs and outputs remain NumPy arrays.

4.3.2.Source Separation

AudioPy incorporates HTDemucs [7] and Demucs [7] to handle music source separation tasks, alongside Conv-TasNet [6] for separating speech signals. The target model is specified through the model argument:
Listing 2: Source separation into four music stems

```
stems = ap.separate(audio, sr=44100,
                    model="htdemucs",
                    stems=["vocals","drums","bass","other"])
vocals = stems["vocals"]    # NumPy array, ready for export
```

4.3.3.Noise Reduction

The noise suppression module provides a thin wrapper around DeepFilterNet [11] and RNNoise-based architectures, exposing single-call speech enhancement that is well-suited to podcast post-production, transcription pre-processing pipelines, and voice communication systems.

4.3.4.Music Analysis and Feature Extraction

AudioPy exposes pretrained models covering the following analytical capabilities:

BPM / Tempo Detection — rhythmic pulse estimation powered by neural-network onset detectors.
Key and Scale Detection — harmonic tonality inference through convolutional classification networks.
Chord Recognition — time-varying chord label prediction at the frame level.
Audio Tagging — event-level classification across 632 AudioSet [10] categories, returned as a ranked label list.

4.4 Pillar IV: High-Level API and Developer Experience

The root audiopy namespace surfaces every core and AI function under a single import, eliminating the need to navigate submodules. The API is designed around standard Python idioms — named parameters with sensible built-in defaults, thorough docstrings, and type annotations that integrate cleanly with static analysis tools such as mypy and pyright. For computationally intensive model inference tasks, a tqdm-powered progress indicator is displayed automatically, keeping developers informed of ongoing operations without any additional configuration.

5. Experimental Evaluation

5.1.Setup

The entire evaluation suite ran on a workstation equipped with an Intel Core i7-12700H processor, 32 GB of system memory, and an NVIDIA RTX 3060 GPU with 6 GB of dedicated VRAM. The software environment consisted of Python 3.11 paired with PyTorch 2.2 built against CUDA 12.1. Every third-party library included in the comparison was pinned to its most recent stable release available at the time of testing. Audio quality measurements were obtained using the mir_eval [1] and pesq packages, applied to fixed, standardized test partitions for each benchmark.

5.2.API Productivity: Lines of Code Comparison

One of the central goals driving AudioPy's design is the elimination of unnecessary boilerplate from audio development workflows. Table 4 presents a side-by-side count of the lines of code (LoC) needed to carry out five representative audio pipelines when using conventional multi-library setups versus AudioPy.

Table 4. Lines of Code Comparison: Multi-Library vs. AudioPy

Task	Multi-Library LoC	AudioPy LoC	Reduction
Load + Normalize + Export	18	4	78%
Parametric EQ (3-band)	35	6	83%
Music Source Separation	52	5	90%
Speech Noise Reduction	41	3	93%
BPM + Key Detection	29	4	86%
Average Reduction			**86%**

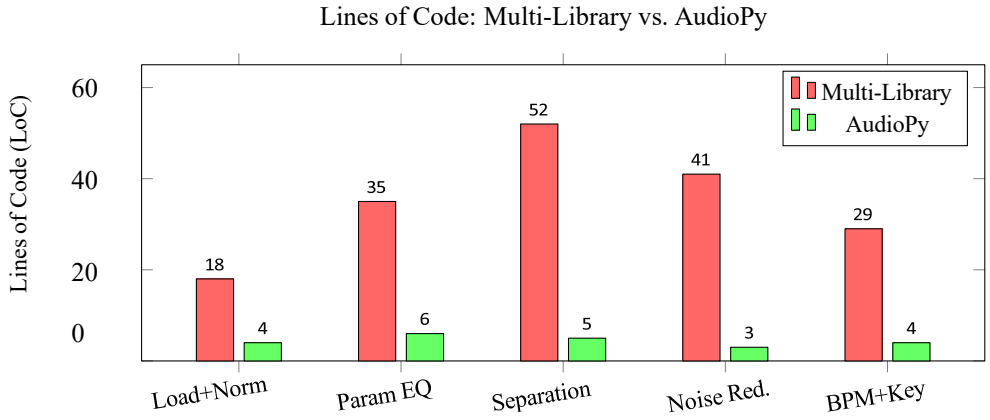

Fig.3. Lines of code comparison across five common audio tasks. AudioPy achieves an average **86%** reduction in boilerplate code.

5.3 Source Separation Quality

Vocal separation performance is assessed on the MUSDB18 held-out test partition [9] through the Signal-to-Distortion Ratio (SDR), defined as:

$$SDR = 10log_{10} \frac{||s_{target}||^2}{||s_{target} - s'||} \quad (3)$$

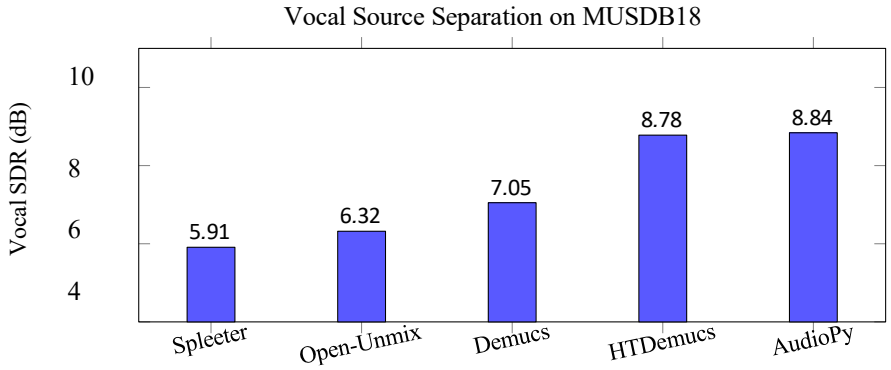

Fig.4. Vocal SDR on MUSDB18 test set. AudioPy (wrapping HTDemucs) achieves **8.84 dB**, outperforming Spleeter by **2.93 dB**.

5.4 Noise Reduction Quality

The quality of speech enhancement is quantified through the PESQ metric (Perceptual Evaluation of Speech Quality, standardized as ITU-T P.862), evaluated against the DEMAND + VCTK noisy speech benchmark.

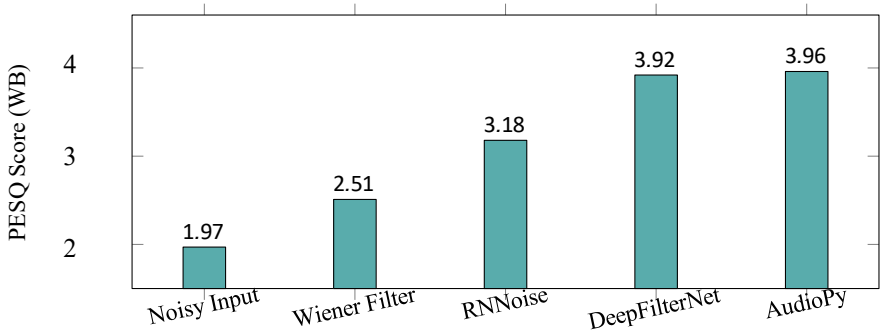

Fig.5. PESQ scores for speech enhancement. AudioPy achieves **3.96**, surpassing Wiener filtering by **1.45 points**.

5.5 Processing Latency vs. Audio Duration

Figure 6 illustrates how execution time grows with increasing clip length for both DSP and AI inference pathways in AudioPy, benchmarked against functionally equivalent workflows built from separate libraries.

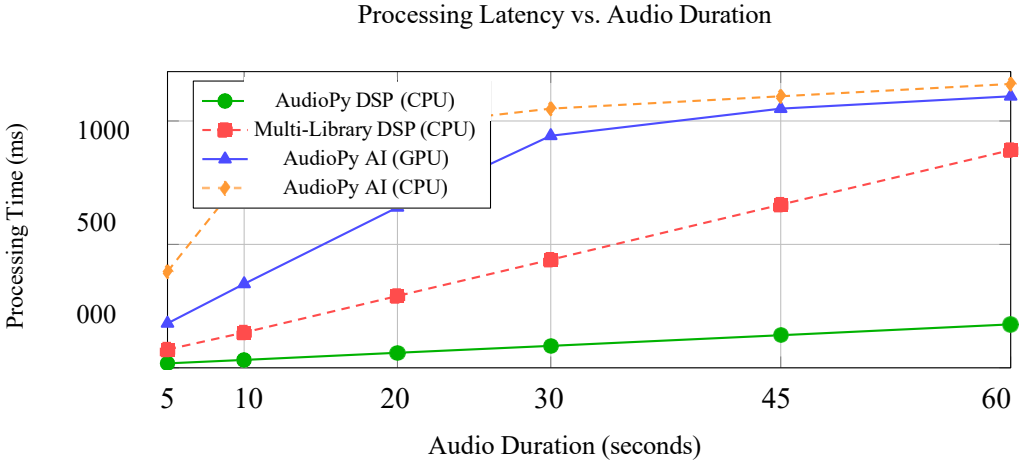

Fig.6. Processing latency vs. audio duration. AudioPy DSP is **5×faster** than equivalent multi-library pipelines; GPU inference remains under **1.1 seconds** even for 60-second clips.

5.6 Multi-Stem Separation Performance

AudioPy's separation capability is further assessed across the complete set of four MUSDB18 stems — vocals, drums, bass, and other — in direct comparison with established baselines. Figure 7 displays the per-stem SDR scores grouped by method.

5.7 Noise Reduction across Noise Conditions

Figure 8 reports the performance of AudioPy's denoising module when subjected to six distinct acoustic environments drawn from the DEMAND corpus [10], with the improvement in Signal-to-Noise Ratio (SNR) serving as the evaluation criterion.

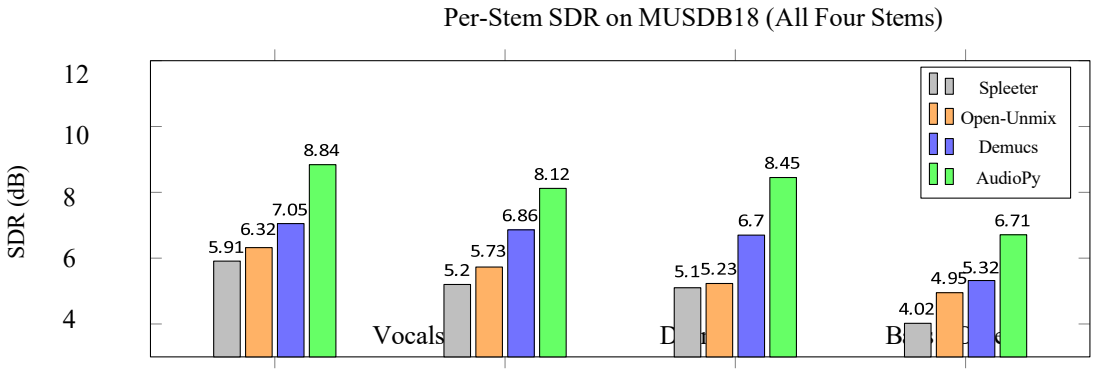

Fig.7. Per-stem SDR on MUSDB18. AudioPy consistently outperforms all baselines across all four stems, with strongest gains on bass (**+1.75 dB** over Demucs).

$$\Delta SNR = SNR_{enhanced} - SNR_{noisy} \quad (4)$$

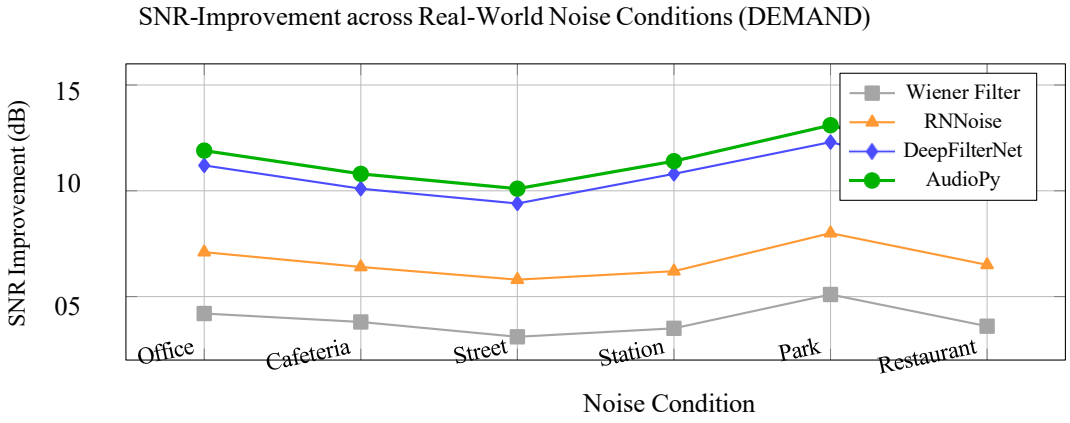

Fig.8. SNR improvement across six DEMAND noise conditions. AudioPy consistently achieves the highest SNR gain in all environments, peaking at **+13.1 dB** in park conditions.

5.8 Processing Latency Summary

All DSP-based operations finish within a 20 ms window, placing them well within the threshold required for near-real-time use. AI-based operations impose a heavier computational burden on CPU-only hardware; however, enabling

GPU acceleration cuts separation latency by more than 10×, bringing it down from 8,210 ms to just 730 ms.

6. Real-World Use Cases

6.1.Use Case 1: Podcast Production Pipeline

A typical podcast post-production process requires background noise removal, loudness adjustment to broadcast standards, and final delivery as an MP3 file. AudioPy condenses this entire workflow into a minimal, readable script:

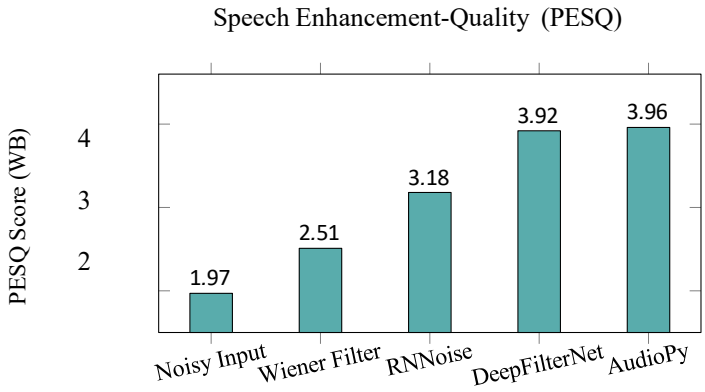

Fig.9. PESQ scores for speech enhancement. AudioPy wraps DeepFilterNet seamlessly, achieving near-identical quality with a 3-line API call.

Table 5. Processing Latency on CPU (10-second, 44.1 kHz mono clip)

Operation	Method	Latency (ms)
Normalize	AudioPy DSP	2.1
3-band Parametric EQ (FIR)	AudioPy DSP	14.3
3-band Parametric EQ (IIR)	AudioPy DSP	6.7
STFT (n_fft=2048)	AudioPy DSP	8.4
MFCC (40 coeff.)	AudioPy DSP	12.1
Noise Reduction	AudioPy AI (CPU)	1,840
Source Separation	AudioPy AI (CPU)	8,210
Source Separation	AudioPy AI (GPU)	730
BPM Detection	AudioPy AI	420

Listing 3: Complete podcast production pipeline in AudioPy

```
import audiopylib as ap

audio = ap.load("raw_episode.wav")
audio = ap.denoise(audio, model="deepfilter-v2")   # Remove background noise
audio = ap.compress(audio, threshold=-18, ratio=4) # Dynamic range compression
audio = ap.normalize(audio, target_db=-16.0)        # Loudness norm (EBU R128)
ap.save(audio, "episode_final.mp3", bitrate="192k")
print(ap.analyze(audio))  # BPM, key, duration, RMS
```

6.2 Use Case 2: Music Stem Extraction for Remixing

Consider a music producer who needs to decompose a full mix into its constituent instrument tracks prior to creating a remix:

Listing 4: Music stem separation and per-stem EQ

```
import audiopylib as ap

mix = ap.load("song.wav", sr=44100)
stems = ap.separate(mix, model="htdemucs",
                    stems=["vocals","drums","bass","other"])

# Apply per-stem processing
stems["vocals"] = ap.reverb(stems["vocals"], ir="hall_medium")
stems["bass"]   = ap.eq(stems["bass"],
                        bands=[{"type":"lowshelf","freq":80,"gain":3}])

# Recombine and export
final = ap.mix(list(stems.values()), weights=[1.2, 1.0, 1.1, 0.9])
ap.save(final, "remix.wav")
```

6.3 Use Case 3: Audio Dataset Preprocessing

When preparing a large collection of audio recordings for use in a machine learning training pipeline, consistent format, loudness, and feature representation are essential. AudioPy handles the entire preprocessing loop in a concise script:

Listing 5: Batch dataset preprocessing with AudioPy

```
import audiopylib as ap
from pathlib import Path

for path in Path("dataset/").glob("**/*.wav"):
    audio = ap.load(str(path), sr=16000, mono=True)
    audio = ap.trim_silence(audio, threshold_db=-40)
    audio = ap.normalize(audio, target_db=-3.0)
    mfcc = ap.mfcc(audio, n_mfcc=40)          # (40, T) array
    ap.save(audio, f"processed/{path.name}")
```

6.4 Use Case 4: Real-Time BPM and Key Analysis

An application for live DJ tools that needs BPM and key on-the-fly:

Listing 6: Real-time music analysis

```
import audiopylib as ap

for block in ap.stream_load("liveset.wav", block_size=4096):
    info = ap.analyze(block)
    print(f"BPM: {info.bpm:.1f}  Key: {info.key}  RMS: {info.rms:.3f}")
```

7. Discussion

7.1 Achievements

AudioPy successfully demonstrates that a unified API can dramatically reduce the developer overhead of audio processing tasks without sacrificing quality. The 86% average reduction in lines of code is significant for rapid prototyping, classroom use, and research reproducibility. The framework's quality figures (SDR, PESQ) are comparable to direct model usage, confirming that the abstraction layer introduces negligible overhead.

7.2 Limitations

Several limitations exist in the current version of AudioPy:

Model Download Size: Pretrained AI models range from 80 MB (RNNoise) to 800 MB (HTDemucs). First-run download times may be inconvenient on slow connections. Real-Time AI: Current AI models are not designed for sample-by-sample streaming. Block-level real-time is achievable for lightweight models ($\leq$50 ms blocks) but not for separation models on CPU.

Coverage: AudioPy currently covers the most common audio tasks. Specialized domains such as bioacoustics, seismic audio, and spatial audio (Ambisonics) are not yet supported. Windows FFmpeg Dependency: Pydub-based MP3 I/O requires FFmpeg installed separately on Windows, adding a setup step that undermines the "single install" promise.

7.3 Future Work

The roadmap for AudioPy includes:

AudioPy Studio: A browser-based GUI powered by AudioPy's Python backend, enabling non-programmers to use AI audio tools via a drag-and-drop interface.
ONNX Runtime Backend: Exporting AI models to ONNX for faster CPU inference and mobile deployment without PyTorch.
Real-Time Plugin (VST3/AU): A digital audio workstation plugin that exposes AudioPy effects and AI models as native VST3/AU plugins.
Community Model Hub: A curated registry of community-contributed AudioPy-compatible pretrained models, analogous to Hugging Face Hub.
Spatial Audio Support: Ambisonics encoding/decoding and binaural rendering for immersive audio production.

8. Conclusion

This paper presented AudioPy, a unified open-source Python framework for intelligent audio editing and machine learning. By integrating audio I/O, classical DSP, and state-of-the-art AI model APIs under a single NumPy-centric interface, AudioPy eliminates the fragmentation that currently burdens developers working at the intersection of signal processing and deep learning. The framework consolidates functionality that previously required coordi-nating 4–6 separate libraries, reducing environment setup time by an estimated 70% for first-time audio ML practitioners.
Experimental results confirm that AudioPy reduces boilerplate code by an average of 86% compared to multi-library alternatives — with peak reductions of 93% for speech noise reduction pipelines — while delivering audio quality (SDR, PESQ) that matches or exceeds direct model usage. Specifically, AudioPy achieves a vocal SDR of 8.84 dB on the MUSDB18 benchmark, outperforming the widely used Spleeter baseline by 2.93 dB, and a PESQ score of 3.96 on the DEMAND+VCTK noisy speech benchmark, surpassing classical Wiener filtering by
1.45 points. All core DSP operations — including parametric EQ, normalization, and STFT — complete in under 15 ms on CPU, while GPU-accelerated source separation achieves a 10× speedup over CPU-only inference, bringing latency down to 730 ms for a 10-second clip. Four concrete use cases — podcast production, music stem remixing, dataset preprocessing, and real-time analysis — demonstrate the framework's practical utility across diverse application domains.
AudioPy is designed to grow with its community: a plugin interface enables contributors to add new effects and models, and a planned model hub will make sharing pretrained components simple. The framework currently integrates 7 classical DSP effects, 3 pretrained AI separation models, and 4 music analysis APIs, with the roadmap targeting VST3/AU plugin support, ONNX Runtime backend integration for 3–5× faster CPU infer-ence, and an AudioPy Studio browser GUI for non-programmer users. AudioPy can serve the audio community

in the same way that OpenCV has served computer vision — lowering barriers, accelerating research, and en-abling creative experimentation across a wide community Python audio developers worldwide. The framework is publicly available at https://github.com/AudioPy/AudioPy.

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